بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

LIVE ONLINE COURSE | COHORT FALL 2026

AI & Machine Learning for Civil Engineers



SESSION 01 · PYTHON AS A TOOL, NOT A BARRIER

From Engineering Problems to Data & Machine Learning

Python & Google Colab: a practical, no-installation starting point.

Dr. Abdul Waris
Research Associate · Universität Siegen, Germany

Technology partner QUDELTA: STUDIOS

WELCOME TO THE COHORT

Machine learning, taught the way engineers think.

8
live, instructor-led sessions

90 min
each, about half of it coding together

4 weeks
from first cell to uncertainty analysis

8
hands-on Colab notebooks, one per session

1
real engineering dataset in every session

IN THE ROOM

B.Tech students | M.Tech students | Ph.D. scholars | Faculty | Practising engineers | **No coding experience needed**



YOUR INSTRUCTOR

Dr. Abdul Waris

Research Associate · Universität Siegen, Germany

Ph.D. in Geotechnical Engineering
Indian Institute of Technology (IIT) Hyderabad

Former Prime Minister's Research Fellow
PMRF, Government of India

Research in explainable ML & reliability
The basis for sessions 07 and 08 of this course

Invited talks
Oregon State University · DY Patil · NIT Srinagar

THE COURSE AT A GLANCE

Eight sessions. One engineering workflow.

PART 1 · FOUNDATIONS

01 **From Engineering Problems to Python** TODAY
AI/ML basics, Google Colab, NumPy & pandas

02 **From Raw Engineering Data to an ML-Ready Dataset**
EDA, data quality, outliers, train/test split, leakage

03 **Your First Engineering Prediction Model**
Linear regression, error metrics, cross-validation

04 **Nonlinear Problems: Trees & Ensembles**
Decision trees, Random Forest, boosting

PART 2 · PREDICTION TO DECISION

05 **From Prediction to Decision: Classification**
Logistic regression, confusion matrix, precision & recall

06 **Artificial Neural Networks**
Architecture, activations, training, overfitting

07 **Can We Trust Our ML Model?**
Feature importance, SHAP, engineering judgement

08 **From Prediction to Probability**
Monte Carlo, ML surrogates, probability of failure

HOW EVERY SESSION RUNS

Concept, then code, then judgement.

Concept & engineering context
Why the method exists

Live code-along in Colab
Real civil engineering data

Interpretation
Does it make sense?

Q&A
Your questions

0 min

90 min

One notebook per session

Structured Colab notebooks you keep and revisit.

Real engineering data

Materials, structures, energy, transport, water and geotechnics.

Recordings

Shared after each session; available until two weeks after the course ends.

Engineering judgement

Validation, limitations and physical plausibility, not just accuracy.

TODAY · SESSION 01

From engineering problems to Python

01 **From engineering problems to machine learning**
AI, ML and DL · supervised and unsupervised learning · one workflow

02 **Python & Google Colab**
Why Python · what a notebook is · Colab vs alternatives · our toolkit

03 **Hands-on: our first cell**
Open the notebook · run code · read the output like an engineer

BY THE END OF TODAY

- ✓ Place AI, ML and DL on one map
- ✓ Tell a regression problem from a classification one
- ✓ Describe the ML workflow for a civil engineering problem
- ✓ Open, run and save a Colab notebook

01

PART 01

From engineering problems to machine learning

a A problem we already know

b AI, ML and deep learning

c Supervised & unsupervised

d One ML workflow

START WITH AN ENGINEERING PROBLEM

Can we predict concrete strength from its mix and age?

Cement	Water	Fly ash	Slag
Aggregates	Super-plasticizer	Age	mix + age

↓ WE MEASURE A RESPONSE

Compressive strength MPa

We already understand this problem. We have the ingredients, we have age, and we measure a response.

THE QUESTION IS

Can we use historical examples to learn the relationship?

CIVIL ENGINEERS ALREADY BUILD MODELS

ML is another engineering modelling tool.

Theory & mechanics	first principles
Empirical correlations	fitted by us
Numerical models	FEM · FDM
Laboratory / field data	measured
Data-driven models	learned from data

We already build models. We use equations, correlations, numerical models, experiments and design standards.

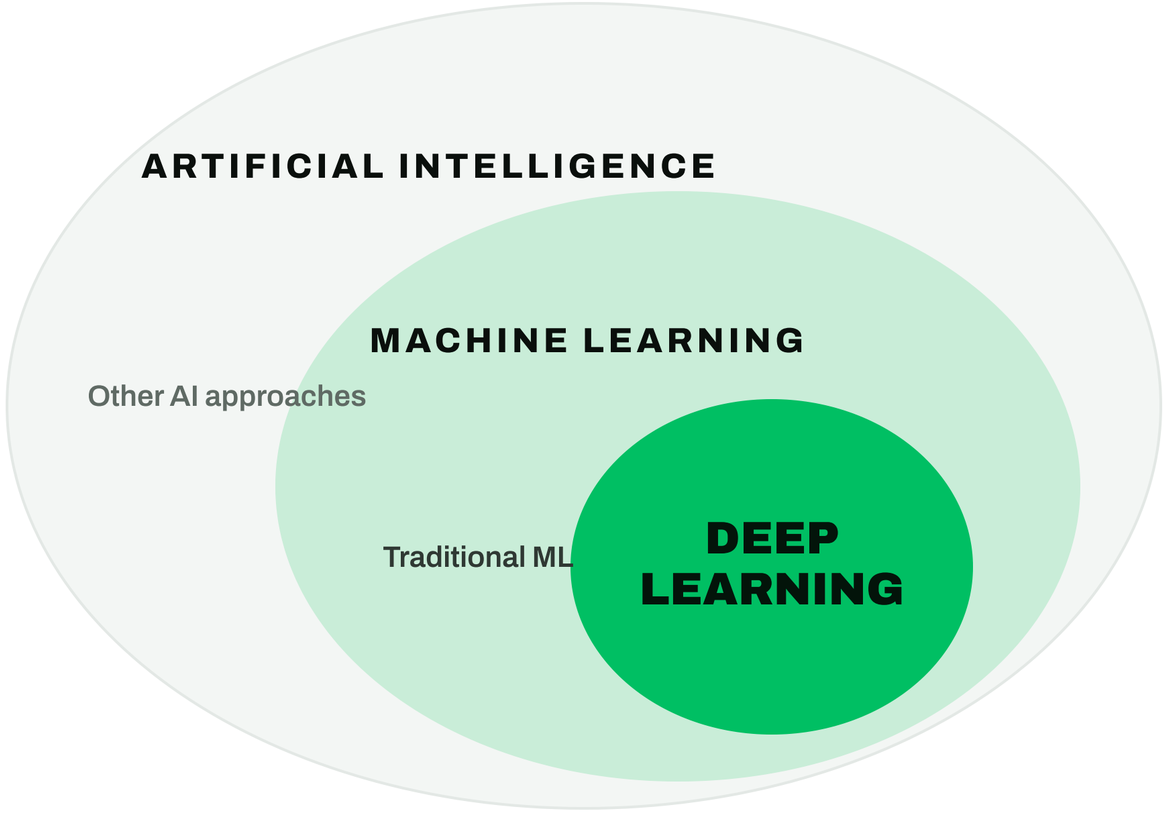
Machine learning doesn't replace engineering thinking. It gives us another way to represent relationships **when we have sufficient data.**

THE DIFFERENCE

Instead of prescribing the relationship ourselves, we let an algorithm learn it from examples in data.

THE VOCABULARY

Artificial intelligence is the broad idea.



Artificial Intelligence
The broader field of building systems that perform tasks associated with human intelligence.

Machine Learning
A subset of AI: systems that learn patterns from data.

Deep Learning
A family of ML methods based on multi-layer neural networks.

Our focus: ML · deep learning in session 06

ONE ENGINEERING EXAMPLE · STRUCTURAL INSPECTION

Same engineering world. Different levels.

LEVEL 1

Artificial Intelligence

Can a system assist an engineer in deciding whether a structure requires attention?

AI = the broader goal

LEVEL 2

Machine Learning

Can we learn from historical inspection data to classify defects?

ML = learning from data

LEVEL 3

Deep Learning

Can a neural network learn complex patterns directly from images of structural defects?

DL = neural-network-based ML

MACHINE LEARNING, DEFINED

> **Field of study that gives computers the ability to learn without being explicitly programmed.**

Arthur Samuel · 1959

TWO FAMILIES OF ALGORITHMS

- Supervised learning
- Unsupervised learning

SUPERVISED LEARNING · EXAMPLES

Input X → output Y

INPUT (X)		OUTPUT (Y)	APPLICATION
Email	→	`spam? (0/1)`	Spam filtering
Audio	→	`text transcript`	Speech recognition
English	→	`Arabic, Urdu, Spanish…`	Machine translation
Ad, user info	→	`click? (0/1)`	Online advertising
Mix design + age	→	`strength (MPa)`	**Civil engineering: our course**

Examples adapted from DeepLearning.AI. Phoon, K. K., Ching, J. & Cao, Z. (2022). Unpacking data-centric geotechnics. *Underground Space* 7(6), 967–989. doi:10.1016/j.undsp.2022.04.001

SUPERVISED LEARNING · LEARNS FROM THE “RIGHT ANSWERS”

Two kinds of supervised problems

REGRESSION

Predict a number

from infinitely many possible outputs.

CIVIL ENGINEERING EXAMPLES

- Concrete strength
- Settlement
- Traffic volume
- Energy demand
- Factor of safety

CLASSIFICATION

Predict a category

from a small number of possible outcomes.

CIVIL ENGINEERING EXAMPLES

- Stable / unstable
- Defect type
- Soil class
- Damage category

The target determines the ML problem.

SUPERVISED LEARNING · WHAT THE MODEL SEES

Fit a curve, or draw a boundary

REGRESSION

Response *y* against a feature *x*



CLASSIFICATION

Two features, two classes

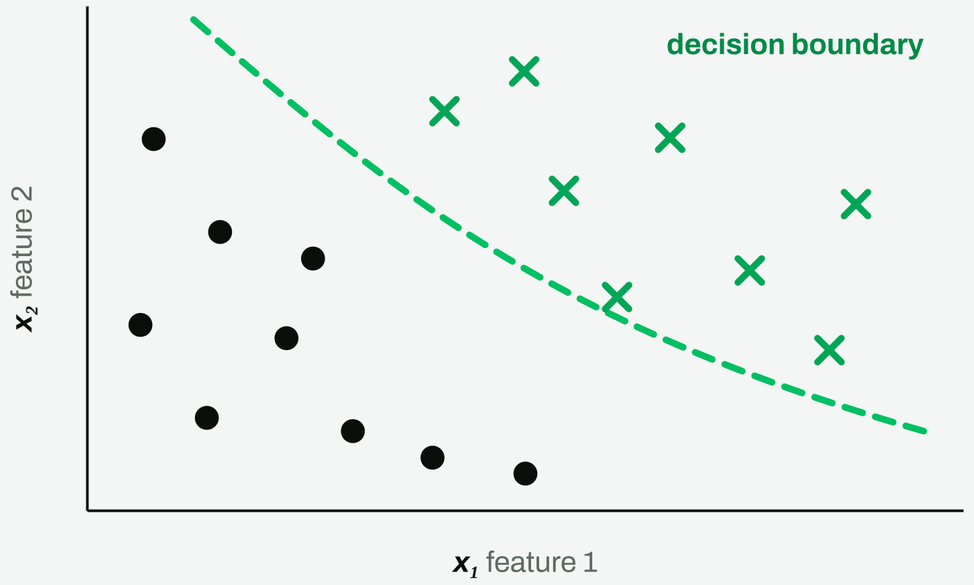
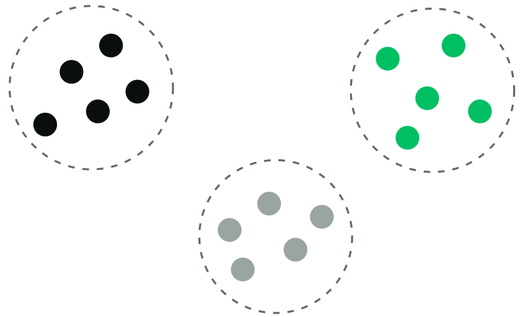


Illustration adapted from DeepLearning.AI.

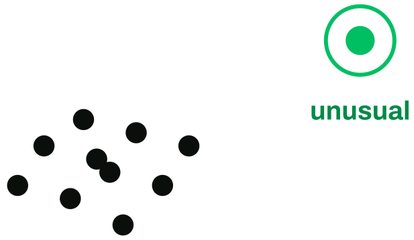
UNSUPERVISED LEARNING · NO LABELS

Only inputs X. The algorithm finds the structure.



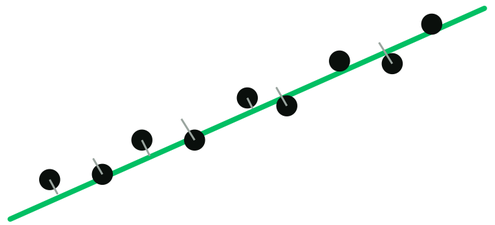
Clustering

Group similar data points together.



Anomaly detection

Find unusual data points.



Dimensionality reduction

Compress data using fewer variables.

DeepLearning.AI. Phoon, K. K., Ching, J. & Cao, Z. (2022). Unpacking data-centric geotechnics. *Underground Space* 7(6), 967–989. doi:10.1016/j.undsp.2022.04.001

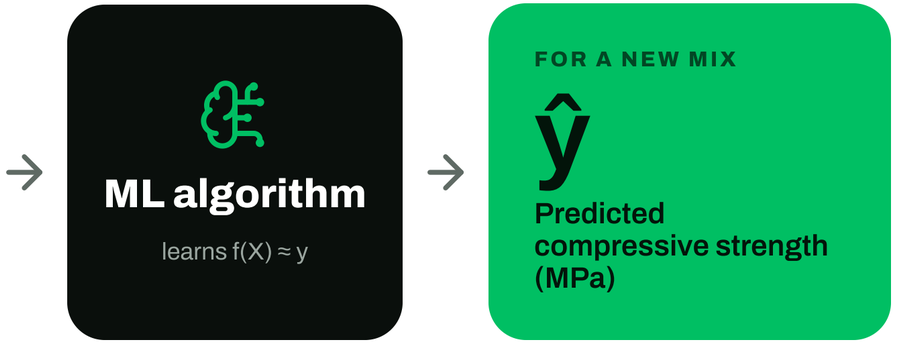
THE PICTURE THAT FOLLOWS US ALL COURSE

X is what we give the model. y is what we want.

INPUTS (X) **OUTPUT (Y)**

Cement	Slag	Fly ash	Water	SP	Coarse agg.	Fine agg.	Age (d)	Strength MPa
540.0	0.0	0.0	162	2.5	1040	676	28	79.99
540.0	0.0	0.0	162	2.5	1055	676	28	61.89
332.5	142.5	0.0	228	0.0	932	594	270	40.27
332.5	142.5	0.0	228	0.0	932	594	365	41.05

… 1,030 mixes in total · quantities in kg/m³



Data: Concrete Compressive Strength dataset, UCI Machine Learning Repository (Yeh, 1998).

THE SIGNATURE OF THIS COURSE

One ML workflow. Many engineering problems.

01
Engineering problem
What do we need to estimate?

02
Data
Measurements, simulations or records

03
Python
Inspect, prepare, analyse

04
ML model
Learn patterns from examples

05
Decision
Interpret with domain knowledge

THE PROBLEM CHANGES. THE DATA CHANGES. THE WORKFLOW SURVIVES.

Concrete	→	Strength	→ Regression
Buildings	→	Energy load	→ Regression
Steel inspection	→	Fault class	→ Classification
Structural health	→	Stability class	→ Classification

02

PART 02

Python & Google Colab

- a Why Python
- b Google Colab & notebooks
- c Colab vs Anaconda vs MATLAB
- d Our Python toolkit

PYTHON

Why are we using Python?

Open-source

No paid language licence is required.

Easy to learn

Readable syntax that is close to plain English.

Scientific ecosystem

NumPy, pandas, Matplotlib, scikit-learn and more.

Useful across disciplines

Data analysis, visualisation, ML and scientific computing.



OUR PRINCIPLE

We are not learning Python for its own sake.

We learn enough Python to ask better engineering questions and build data-driven models.

A SHORT BREAK · WHAT IS IT WITH THE NAME?

Python is named after a comedy show, not a snake.

The name doesn’t come from the snake, even though the logo and community often use snake imagery.

Python was named after the British comedy group **“Monty Python’s Flying Circus”**, a favourite of Python’s creator, **Guido van Rossum**.

He wanted a name that was short, unique and a little mysterious. “Python” felt just right.

FUN FACT



Python, 1991 · Guido van Rossum

WHERE PYTHON FITS

Python supports the engineering workflow

01
Engineering problem
What do we need to estimate?

02
Data
Measurements, simulations or records

03
Python
Inspect, prepare, analyse

04
ML model
Learn patterns from examples

05
Decision
Interpret with domain knowledge

PYTHON SUPPLIES

The workflow: inspect, prepare, model.

ENGINEERING KNOWLEDGE SUPPLIES

The meaning: is this result plausible?

OUR ENVIRONMENT

What is Google Colab?

Hosted notebook environment
Write and run Python in a web browser.

Interactive
Combine code, text, equations and plots.

Low setup burden
No local Python installation is needed to begin.

Shareable
Notebooks can be shared and revisited online.



FOR THIS WORKSHOP

Open the link → run a cell → inspect the output.

We focus on engineering ideas rather than installation and environment setup.

`colab.research.google.com`

ANATOMY OF A NOTEBOOK

What is a notebook?

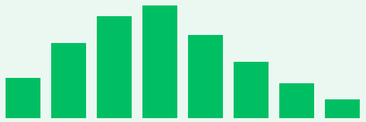
session01_first_steps.ipynb

Water–cement ratio

Assumption: mix quantities in kg/m³. Lower w/c generally means higher strength.

```
import pandas as pd
df = pd.read_csv("concrete.csv")
df["w_c"] = df.water / df.cement # new feature
```

```
mean w/c = 0.54
```



Text / Markdown cells

Explain the engineering question, assumptions, equations and interpretation.

Code cells

Run Python instructions and see the result beneath the cell.

Outputs

Numbers, tables, charts and model diagnostics in one place.

A notebook is a record of code, results and reasoning, not just a place to type code.

THE CHOICE WE MADE

Why Google Colab for this workshop?

No installation
Start in a browser.

Shared workflow
Everyone follows the same notebook sequence.

Easy to share
Links make distributing notebooks straightforward.

Live demonstrations
Run cells and inspect results together.

Engineering-focused
More time for data and interpretation.

Beginner-friendly
No environment management at the outset.

We choose Colab to reduce setup friction, not because it is best for every project.

ALTERNATIVES

Google Colab, Anaconda and MATLAB

FEATURE	Google Colab	Anaconda + Jupyter	MATLAB
Initial setup	**Browser; no local install to start**	Install and manage a local distribution	Installation and licence access may be required
Language	**Python**	Python (with R tools available)	MATLAB
Packages / tools	**Common libraries available; some setup may be needed**	Flexible packages and environments	MATLAB toolboxes and add-ons
Best fit here	**Shared, low-friction workshop**	Local work and controlled environments	Existing MATLAB workflows

All three support technical work; this workshop prioritises accessibility and shared hands-on learning.

GOOD TO KNOW

What should we know about Colab?

Internet access is needed
Colab runs in the browser.

Sessions are temporary
Save notebooks and important outputs.

Files and packages may need setup
Upload data or mount cloud storage when required.

Compute resources can vary
Hardware and usage limits are not guaranteed.

Protect sensitive data
Follow institutional data-protection rules before uploading data.

Use Colab as a practical workshop environment, not as a promise of unlimited compute.

TOOLS, INTRODUCED WHEN WE NEED THEM

Our Python toolkit

Python	Language and logic	session 01
NumPy	Numerical arrays and calculations	session 01
pandas	Tables and data preparation	sessions 01–02
Matplotlib	Plots and visual diagnostics	sessions 01–02
scikit-learn	Machine-learning models and evaluation	from session 03

We introduce each tool when an engineering question calls for it.

The goal is not to memorise every function. It is to understand **what each step does** and **whether its result makes engineering sense.**

03

PART 03 · HANDS-ON

Let’s get started with Python in Google Colab

a Open and run the notebook

b Our first cell

c Working with notebooks

HANDS-ON · STEP BY STEP

How to open and run the workshop notebook

1

Open the shared Colab link

colab.research.google.com

2

Sign in to Google if prompted

Any Google account works.

3

Open the provided notebook

One notebook per session.

4

Run a cell

Click ▶ or press `Shift` + `Enter`

5

Read the output before moving on

Does it make sense?

⚠ If a cell fails, read the error and check the preceding steps before rerunning everything.

AN ENGINEERING-FLAVOURED EXAMPLE

Let’s run our first cell

[1] Python 3

```
cement = 350    # kg/m³
water  = 180    # kg/m³
water_cement_ratio = water / cement
print(water_cement_ratio)
```

EXPECTED OUTPUT

```
0.5142857…
```

Who does what?

The computer performs arithmetic; **the engineer interprets what the ratio means.**

Try it

Change the water content. Observe what changes, and what the calculation alone cannot tell you.

GOOD HABITS FROM DAY ONE

Working with workshop notebooks

01 **Open the shared link**
Use the notebook for the relevant session.

02 **Keep your own copy**
File → Save a copy in Drive when appropriate.

03 **Work in sequence**
Run cells from top to bottom unless instructed otherwise.

04 **Keep the analysis reproducible**
Record assumptions and note any changes to evaluation steps.

WORKSHOP NOTEBOOK

Open today’s notebook and make your own copy.

Link shared in the session chat

SUMMARY

Why this setup?

01

Python

An accessible, widely used language for scientific computing and ML.

02

Google Colab

A browser-based notebook environment with minimal setup.

03

Engineering judgement

The goal is meaningful analysis, not merely code that runs.

NEXT · SESSION 02

From Raw Engineering Data to an ML-Ready Dataset

Load a dataset, inspect its structure, and ask our first engineering questions.

Questions?



Dr. Abdul Waris

Research Associate · Universität Siegen, Germany

END OF SESSION 01

Thank you.

See you in session 02: From Raw Engineering Data to an ML-Ready Dataset.

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Research Associate · Universität Siegen, Germany



Course page
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